

Type foundry

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.2 Programming fundamentals — Paper 2, Computational thinking, algorithms and programming
Lesson	Type foundry (lesson 2 of the unit)
Lesson ID	pf-102
Experience ID	pf-102
Plan ID	PLAN-PF-L02
Content version	45c100e2e032
Website	https://revise360.co.uk/experience.html?id=pf-102

Big question, outcomes and success criteria

Big question — Why does a program need to know what type of data it is working with?

Prerequisites

The lesson before this one is pf-101. Pupils should already have finished it: the starter below retrieves from it.

Assessable outcomes

1. **pf-102-o1** — know the data types integer, real, Boolean, character and string
2. **pf-102-o2** — understand why casting is needed and how a value is cast from one type to another
3. **pf-102-o3** — know how the arithmetic, comparison and Boolean operators are used

Vocabulary, resources and preparation

Key vocabulary — Casting, Concatenation

Resources

SLIDES	presentations/PF_L02_TypeFoundry_Lesson.pptx
WORKSHEET	worksheets/PF_L02_TypeFoundry_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=pf-102

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.

3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on PF_L02_TypeFoundry_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Data types.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Data types

- An integer is a whole number, positive or negative, with no decimal part. A real holds a number with a decimal part.
- A Boolean holds only True or False.
- A character is one single symbol; a string is a sequence of characters, and may be empty.

Challenge on this wall: Choose a data type for a pupil's name, their age, their average mark and whether they are present.

2. Casting

- Casting converts a value from one data type to another.
- In the OCR Exam Reference Language int() casts to an integer, float() to a real, str() to a string and bool() to a Boolean.
- Casting is needed because input() always returns a string, and strings cannot be used in arithmetic.

Challenge on this wall: Explain why a program that adds up marks must cast each input before it adds it to the total.

3. Arithmetic operators

- The arithmetic operators are + to add, - to subtract, * to multiply and / to divide.
- In the OCR Exam Reference Language the exponent operator is ^, so 3^2 means 3 to the power 2, which is 9.
- The usual order of operations applies: brackets, then exponents, then multiply and divide, then add and subtract.

Challenge on this wall: Work out $3 + 4 * 2$ and explain the order you used to get your answer.

4. MOD and DIV

- DIV gives the whole number part of a division and throws the remainder away, so $17 \text{ DIV } 5$ is 3.
- MOD gives the remainder left after a whole number division, so $17 \text{ MOD } 5$ is 2.
- MOD 2 is a quick test for an even number, because an even number always leaves a remainder of 0.

Challenge on this wall: Use MOD and DIV to turn 95 seconds into a number of whole minutes and a number of seconds left over.

5. Comparison and Boolean operators

- The comparison operators are ==, !=, <, >, <= and >=, and each one gives True or False.
- AND is True only when both conditions are True. OR is True when at least one condition is True.
- NOT reverses a condition, so NOT True gives False.

Challenge on this wall: Write a condition that is True only when a mark is between 40 and 69, including both ends.

6. String manipulation

- The length of a string is the number of characters in it, counting spaces.
- A substring is a part of a string, taken from a given position. Concatenation joins strings together using +.
- In the OCR Exam Reference Language name.length, name.substring(0, 3), name.upper and name.lower do this work, and ASC() and CHR() convert between a character and its ASCII code.

Challenge on this wall: Describe how you would build a username from the first three letters of a surname and a pupil's year group.

Retrieval and the check for understanding

Starter — A school report stores a pupil's name, their average mark to one decimal place, their form group, and whether they have full attendance. Write down the most suitable data type for each one.

Answer space: 4 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pf-102-s4-c1), so the wording a pupil sees matches the wording you say.

What is the value of $23 \text{ MOD } 4$?

Correct: 3

What a pupil might choose instead

- 5
- 4
- 19

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: 4 goes into 23 five times with 3 left over, and MOD gives that remainder.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	14	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	1	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Data types	2	A real number is different from an integer because...
2	Casting	2	Casting is...
3	Arithmetic operators	2	The exponent operator...
4	MOD and DIV	2	MOD gives... DIV gives...
5	Comparison and Boolean operators	2	AND is True only when...
6	String manipulation	2	Concatenation means...
final	Final challenge: work it out	3	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pf-102-x1	Identify the most suitable data type for each of these values: 17, 3.5, True, "Year 10".	4	worksheet, Exam practice a); deck, Exam practice slide
pf-102-x2	Explain why casting is needed when a program inputs a number and adds it to a running total.	3	worksheet, Exam practice b); deck, Exam practice slide
pf-102-x3	A program stores a length of time in seconds. Describe how MOD and DIV could be used to output that time in whole minutes and remaining seconds.	4	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pf-102.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Which data type should store the price 4.99?** — Integer; Boolean
- **Which data type can hold only two possible values?** — Character; Integer
- **What does casting mean?** — Rounding a number to the nearest whole number; Giving a variable a new name
- **What is the value of `int("25") + 5`?** — "255"; "25" + 5
- **In the OCR Exam Reference Language, what is the value of `5 ^ 2`?** — 10; 7

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- The five data types and when to use each one
- What casting is and why `input()` makes it necessary
- MOD, DIV and the Boolean operators AND, OR and NOT

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is pf-103, Filing room. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

Revise 360 — a product of Revise 360 Ltd, created by Olly Pettitt. An independent resource: not written, endorsed or approved by OCR or any exam board. Classroom use and personal study only.